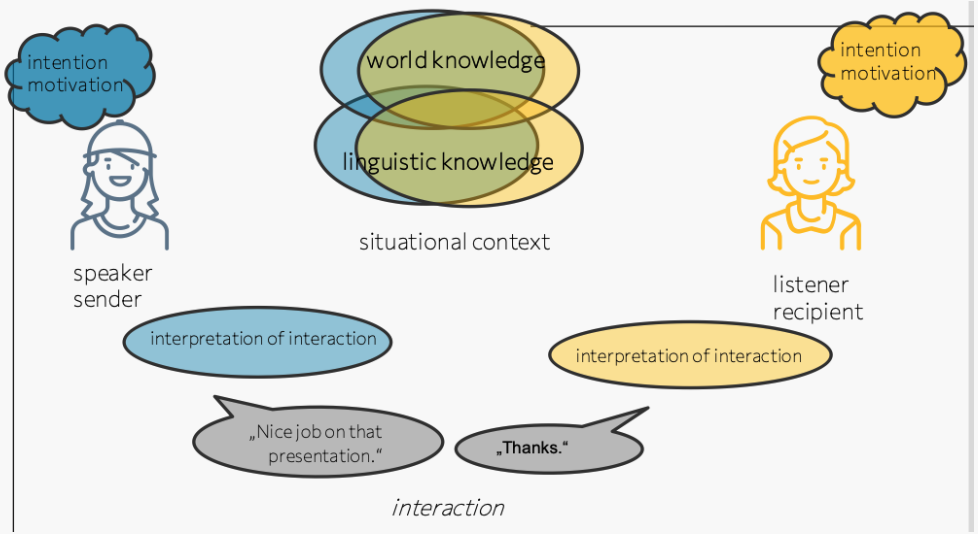
What have we learned?
Context matters



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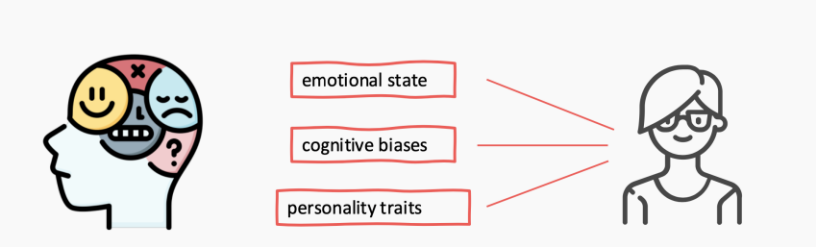
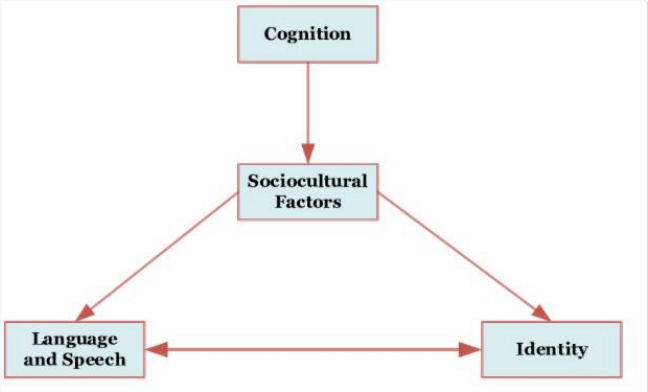
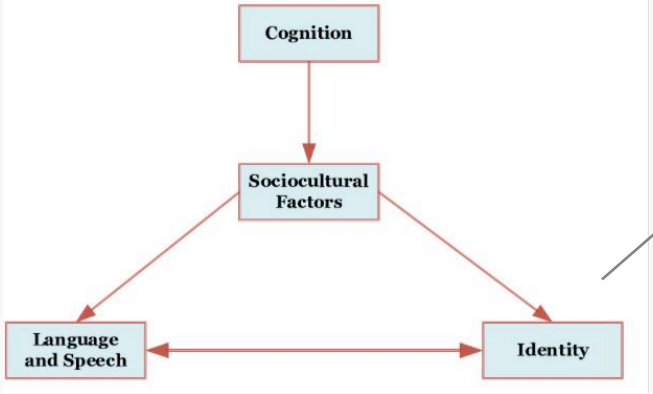
Context matters



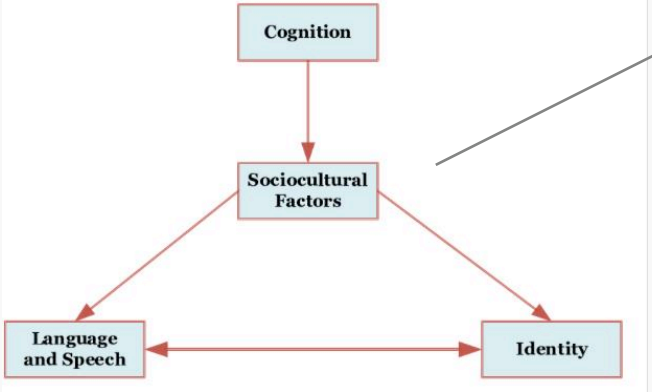


What have we learned?
Context matters

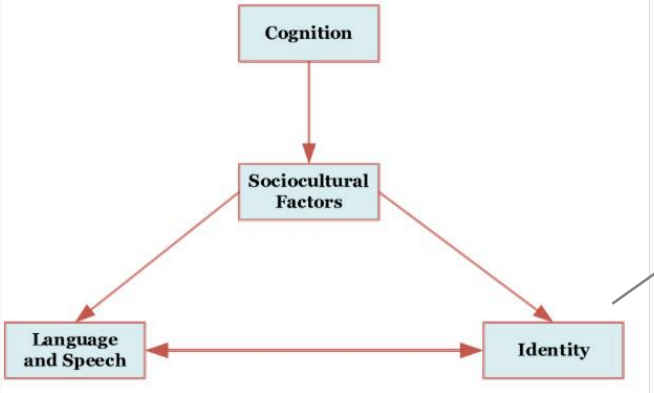


linguistic variance / styles

What have we learned?
Context matters



Context that is missing



in relation to *society*

e.g. deliberation: group dynamics and coorporation in interactions
collective identity through storytelling

Practical Project 2

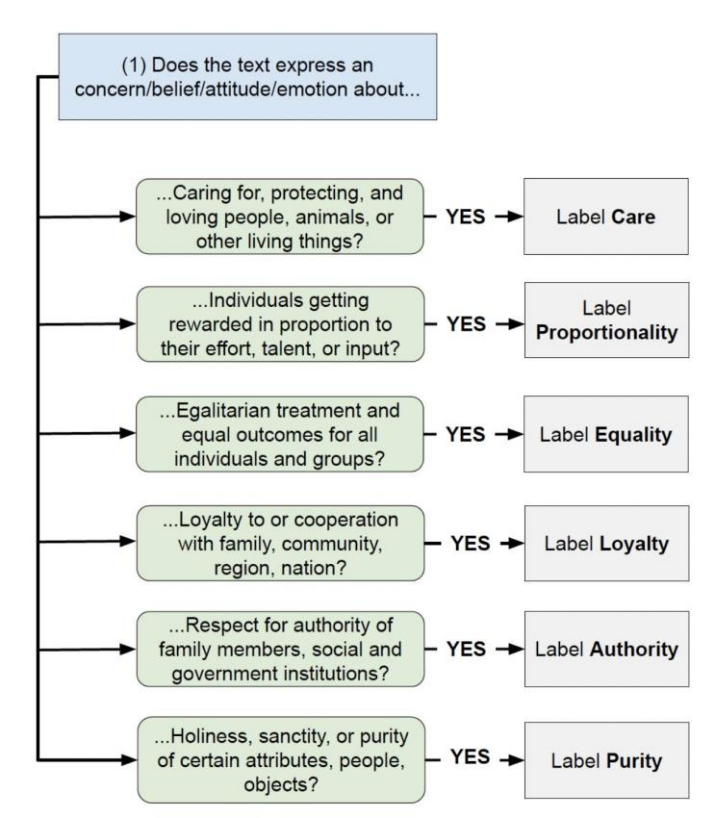
Moral Foundations and Human Values

Moral Foundation Theory

How do we measure it in humans?

CONSTRUCT	DEFINITION	"indicate how well each statement describes you or your opinions."
Care	Protection from suffering	Caring for people who have suffered is an important virtue.
Equality	Equal outcomes for individuals	I believe that everyone should be given the same quantity of resources in life.
Proportionality	Being rewarded proportionally	The effort a worker puts into a job ought to be reflected in the size of a raise they receive.
Authority	Respect for Tradition	I think obedience to parents is an important virtue.
Loyalty	Group Solidarity	Everyone should defend their country, if called upon.
Purity	Harmony of Body and Soul	People should try to use natural medicines rather than chemically identical human-made ones.

Main Dataset: MFRC



paper link:
https://arxiv.org/pdf/2208.05545

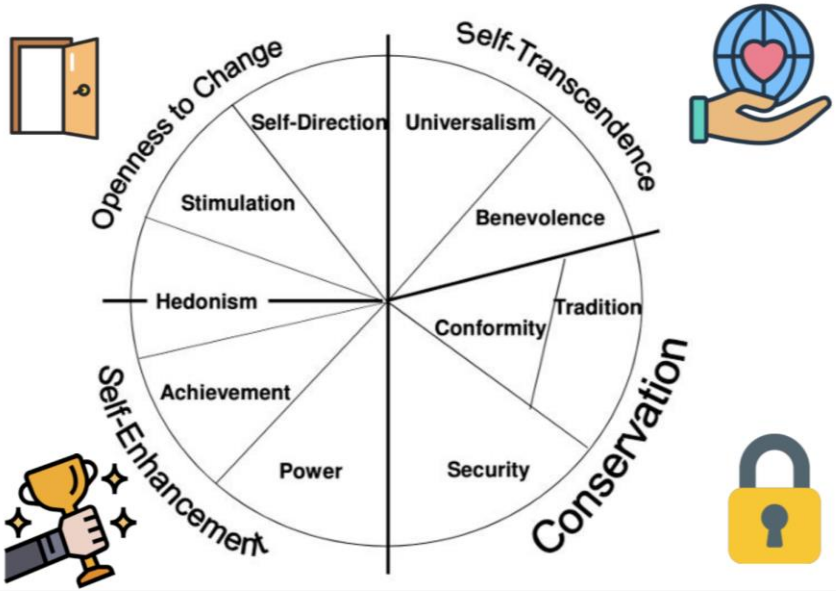
How does the data look like?

ID	text	Equality	Care	Non-Moral	Authority	Thin Morality
3	>Ergo, he supports Macron but doesn't want to say it out loud? Jesus grow a spine you old commie idiot	0	0	1	0	1
4	>He looks exactly the same in Richie Rich as he does as Mr. Parrish in Jumanji. Amazing, really. Almost as though he's the same person.	0	0	2	0	1

location of the data on the server:
/tmp/data_nlpforsocialinteractions/MFRC_nlpforsocialinteractions.csv

Schwartz's Theory of Values

Values are a continuum and may stand in conflict to one another



“Imagine someone is arguing that the South Africa’s COVID-19 lockdown was too strict by saying …”

“The economic ramifications of the lockdown have been huge, and have been felt hardest by those who were already most vulnerable.”

If asked why is that good, might this be their justification?

Because it is good to …
- be rich
- be just
- have a stable society…

Identifying the Human Values behind Arguments
(Kiesel et al, 2022)
https://aclanthology.org/2022.acl-long.306.pdf

How does it look like?

Argument ID	Conclusion	Stance	Premise
A01002	We should ban human cloning	in favor of	we should ban human cloning as it will only cause huge issues when you have a bunch of the same humans running around all acting the same.
A01005	We should ban fast food	in favor of	fast food should be banned because it is really bad for your health and is costly.
A01006	We should end the use of economic sanctions	against	sometimes economic sanctions are the only thing that will get the corrupt governments to take action
A01007	We should abolish capital punishment	against	capital punishment is sometimes the only option to keep criminals from committing more crimes.
A01008	We should ban factory farming	against	factory farming allows for the production of cheap food, which is a necessity for families surviving on a low income.

Self-direction: thought	Self-direction: action	Stimulation	Hedonism	Achievement	Power: dominance	Power: resources	Face	Security: personal
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	1
0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	1

links to the papers:
https://aclanthology.org/2023.semeval-1.313.pdf

location of the data on the server:
/tmp/data_nlpforsocialinteractions/human_values_nlpforsocialinteractions.csv

Project Task Version 1 - modeling

Project Task Version 1 - modeling

using the MFRC, the modeling task is: **does this text express a certain moral foundation or not?** (binary classification for one specific foundation)

- choose *three moral foundations* you want to focus on (three different classification tasks)
- compare two modeling approaches:
  - (1) a feature-based (BoW or with linguistic features)
  - (2) fine-tune some BERT variant
- feature-based: can be BoW, your previous features, some other linguistic features
- dictionaries with moral foundation and value-related terms can also be used, located under /tmp/data_nlpforsocialinteractions
- *RQ1: how well can we model it/ which model is good / feature?*

Project Task Version 1 - modeling

for RQ2, apply the best classifier to either

- (a) a subset of the human values corpus. *investigate co-occurrence patterns of your foundations and human values.* Focus on the human values with the highest co-ocurrences with your foundations.
- *(b)* two subreddits of your choice (subsets). compare the frequency of each foundation? *are there any differences between the communities?* Do some qualitative analysis.

Project Task Version 2 - annotation

can we **simulate moral foundation annotation with an LLM**? How does context influence LLM annotation?

- choose *three moral foundations* you want to focus on (three different classification tasks)

text	subreddit	bucket	annotator	annotation	confidence
TBH Marion Le Pen would be better. Closet fascist vs flamboyant extroverted fascist.	neoliberal	French politics	annotator03	Non-Moral	Somewhat Confident
TBH Marion Le Pen would be better. Closet fascist vs flamboyant extroverted fascist.	neoliberal	French politics	annotator00	Thin Morality	Not Confident

Project Task Version 2 - annotation

can we **simulate moral foundation annotation with an LLM**? How does context influence LLM annotation?

- choose *three moral foundations* you want to focus on (three different classification tasks)
- use the LLM as an annotator, develop a good prompt for the task (e.g. use the one human annotators received).
- investigate socio-demographic prompting:
- RQ: *does annotation context (giving LLM identity-related background about annotator) influence the annotation behavior?*
- does a certain context align better with a certain annotator?

Project Task Version 2 - annotation

- test on yourself, annotate a small subset (50 instances) with the same schema. Does the LLM align with you if you specify, for example your own moral preferences / your age...
- **attention**: critically reflect on socio-demographic prompting (advantages / disadvantages in your report)!

paper about socio-demographic prompting, useful information and prompt templates:
https://aclanthology.org/2025.findings-emnlp.1261.pdf

Project Extension - Modeling

- include more context, a more advanced model or more advanced features
- more in-depth investigation of one of the previous classification tasks (e.g. specific empathy type) or a 4-way classification
- within the subreddits there might be further distictions you want to look at (e.g. advice giving vs advice receiving, the post vs. the responses)

Project Extension - Annotation

- add another layer of annotation to the data to see how it relates to the output variable / the two subreddits
- define your phenomenon and some guidelines. using your guidelines manually annotate a subset of 100 posts
- use LLM as an annotator, how well does it do on your test set?
- annotate a bigger subset with the LLM (2000 instances), fine-tune a BERT on these labels if you want to have a predictive feature-extractor for more data
- RQ: *what is the relationship between your new phenomenon and the target variable (empathy type, community A vs B?*

LLM as an annotator

- provided on the server
- two scripts
- (a) test prompt (send a prompt and an example to the model and test what the model responds)
- (b) once you have a prompt you are happy with run the model on a full dataset (max 2000 instances in one run)
- server: ssh studentaccount@fleckenweihe.ims.uni-stuttgart.de (only use for the purpose of this course!)
- create a folder for yourself (your own data and results in /tmp).
- a very detailed instruction of what to do, how to run and prompt the model is provided on the GitHub in *readme_assignment2.md*

Next Steps

submission deadline: 31st of March, same requirements for the report
teamwork is possible (again, each submits their own report)
you need to get your topic confirmed by me. decision needs to be made by next week:
- short slot to present me what topic you have decided to go for
  next Tuesday: 15:00 – 17:00
  or Wednesday: 09:00 – 11:00
- to book a slot, put your name in the poll: https://dud-poll.inf.tu-dresden.de/Buc8DiDkoQ/
- we can meet online in my webex room: https://unistuttgart.webex.com/meet/neele.falk